

Kernel Span in TextCNN for Seven-Way Harry Potter Page Attribution

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Abstract—We study how the receptive field of a convolutional text classifier affects seven-way attribution of lines (treated as pages) to Harry Potter books. Words are embedded using our prior word2vec encoder with `<pad>` and `<unk>`; the embedding space is extended by one dimension to host a one-hot UNK flag. A TextCNN with parallel 1D convolutions is trained while varying kernel sets. Holding optimisation, data split, and capacity per branch fixed, we find that *short* kernels (3, 4, 5) provide the best average generalisation (mean test accuracy 0.567, macro-F1 0.567; three seeds), whereas wider spans (4, 5, 6), (5, 6, 7) and any set including $k=8$ reduce performance. Figures on the following page summarise the quantitative and qualitative evidence.

I. INTRODUCTION

Convolutional models for text classify sequences by sliding n-gram detectors over embeddings, followed by global max-over-time pooling. We examine whether increasing the context window (kernel span) improves seven-way book attribution. The question is natural for this genre: stylistic cues tend to be short, but longer phrases may also be informative. Our aim is to isolate the effect of kernel span under controlled conditions and to present compact, defensible evidence.

II. METHODOLOGY

Data and preprocessing. We use the seven Harry Potter books; each line is a page. Text is lowercased; punctuation and ebook quotes/dashes are stripped; tokens map to indices with out-of-vocabulary words sent to `<unk>`.

Encoder. We load the prior word2vec encoder, add `<pad>` and `<unk>`, extend the embedding dimension by +1, and set the UNK row's last coordinate to 1; the embedding matrix remains frozen during CNN training.

Model. TextCNN with parallel Conv1d branches for each kernel $k \in K$, ReLU, global max pooling, dropout, and a linear classifier. Branches are *parallel*, not sequential.

Training. 80/10/10 split, Adam ($1r \cdot 10^{-3}$), batch 64, 20–30 epochs; the best-validation checkpoint is restored before test evaluation.

Experimental factors. We vary kernel sets while keeping filters-per-branch $n_f=200$ and dropout 0.5. Seeds {123, 456, 685} are used for robustness. We also inspected a percentile length cap (95%) but retained full lines after observing lower validation and test scores.

Metrics. Test accuracy and macro-F1; confusion matrices and validation curves are included for interpretation.

III. RESULTS

Aggregate performance. Table I reports seed-averaged results; [3, 4, 5] is best on mean, while sets with $k=8$ are consistently worst. The best single run overall is [4, 5, 6] with seed 685 (test 0.5857); however its *mean* is below [3, 4, 5], so we select [3, 4, 5] as our final configuration.

Seed-wise variability. Across all sets the standard deviation of test accuracy is $\approx 0.02-0.03$. This variance is visible in the bar plots with error bars (Fig. 2, Fig. 1).

Teammate ablations. Additional experiments conducted by a teammate include [3, 3, 3] (test ≈ 0.565), [1, 1, 1] (≈ 0.496), an extreme [page, page, page] (≈ 0.276) and a mixed [3, 10, page] (≈ 0.362). These reinforce the trend that short n-grams dominate and that very long kernels behave like brittle templates.

IV. DISCUSSION

The consistent advantage of [3, 4, 5] suggests that short phrases—names, stock expressions, formulaic descriptions—suffice for attribution, while larger windows dilute signal with many low-yield positions. Under global max pooling, longer inputs increase the number of candidate windows, raising the chance of spurious high activations; adding $k=8$ worsens this effect, which we see in both accuracy and macro-F1 (Figs. 2, 1). Validation traces for the best run show early saturation without late-epoch collapse (Fig. 4); since we always reload the best-validation checkpoint, extending training to 30 epochs does not harm final test results. Confusion matrices illustrate where kernels help: the selected [3, 4, 5] setting improves separation for books with distinctive recurring phrases (Fig. 5), while a poor setting [3, 8, 5] exhibits increased confusion among mid-series books (Fig. 6). Finally, qualitative inspection of top-activating n-grams (Fig. 3) shows that filters with $k \in \{3, 4, 5\}$ fire on high-precision cues, aligning with the quantitative advantage.

V. CONCLUSION

Short receptive fields are sufficient and preferable for this task. Parallel kernels [3, 4, 5] yield the most reliable performance; widening to [4, 5, 6] or [5, 6, 7], or including $k=8$, degrades generalisation. Future work may hold total convolutional parameters constant across kernel sets, explore dilated convolutions to expand effective context without very large k , and lightly fine-tune embeddings to probe whether results persist.

TABLE I
KERNEL SETS (MEAN OVER THREE SEEDS).

Kernels	Test Acc	Macro-F1
[3, 4, 5]	0.567	0.567
[4, 5, 6]	0.552	0.559
[5, 6, 7]	0.539	0.545
[3, 4, 8]	0.543	0.548
[8, 4, 5]	0.534	0.546
[3, 8, 5]	0.532	0.546

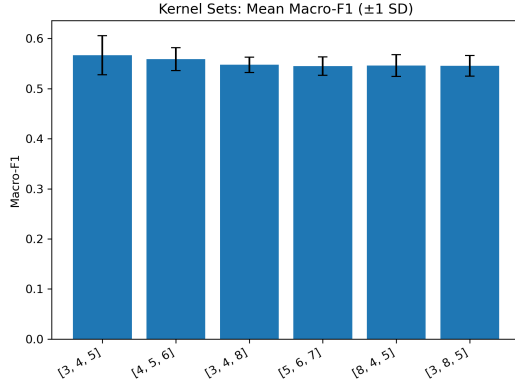


Fig. 1. Macro-F1 by kernel set (mean $\pm$ sd over seeds).

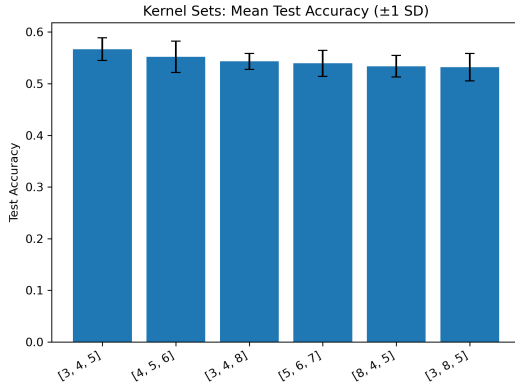


Fig. 2. Test accuracy by kernel set (mean $\pm$ sd over seeds).

Top-k N-grams

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298 'and <unk> for', 'and excitement for', 'and ga...
168 'and the goblet', 'and the goblet', 'and the g...
286 'harry said and', 'harry potter and', 'harry p...
277 'dumbledore <unk> <unk>', 'dumbledore <unk> <u...
269 'of fire <unk>', 'of fire <unk>', 'of fire <un...
293 '<unk> <unk> <unk>', '<unk> <unk> <unk>', '<un...
271 'ticket barrier', 'tibbies snowy', 'be twelve'...
210 'around at <unk>', 'down at ginny', 'down and...
170 '<unk> <unk> <unk>', '<unk> <unk> <unk>', '<un...
171 '<unk> the', 'ollivander his', 'potter <unk> t...
175 'harry potter and', 'harry potter and', 'harry...
267 'potter <unk> the', 'potter and the', 'potter...
266 'looked as though', 'looked as though', 'looke...
177 'than the other', 'than their <unk>', 'than th...
260 'goblet of fire', 'goblet of fire', 'goblet of...
209 'the goblet of', 'the goblet of', 'the goblet...
192 'the goblet of', 'the goblet of', 'the goblet...
249 'harry potter and', 'harry potter and', 'harry...
199 'of fire <unk>', 'of fire <unk>', 'of fire <un...

```

Fig. 3. Top-activating n -grams for a representative filter (teammate ablation).

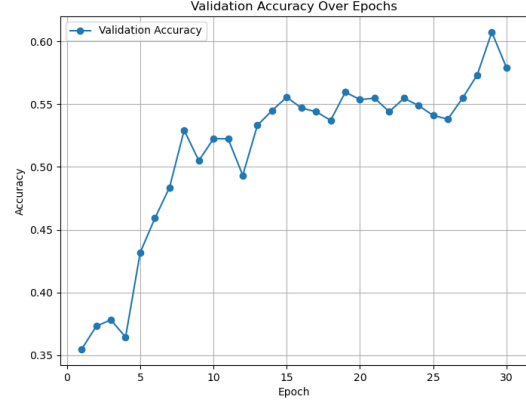


Fig. 4. Validation accuracy over epochs for the best run.

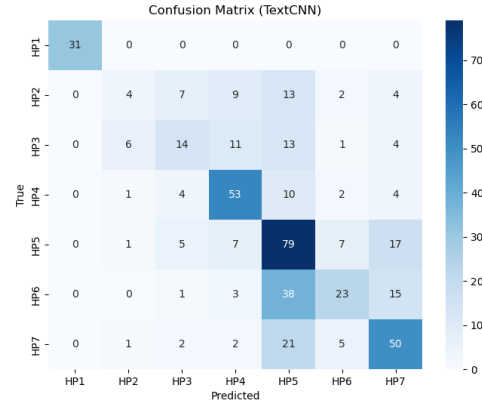


Fig. 5. Confusion matrix for the selected kernel set [3, 4, 5].

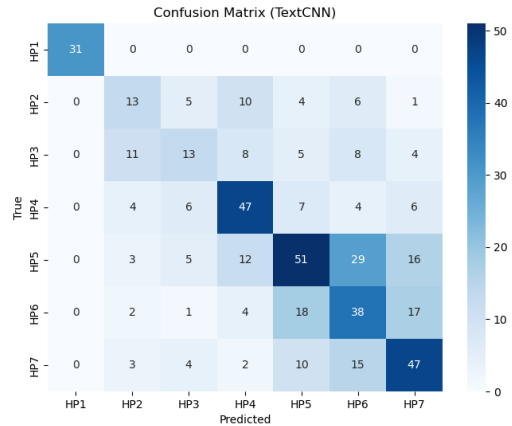


Fig. 6. Confusion matrix for a poor kernel set [3, 8, 5].

CONTRIBUTIONS

Luke Renton implemented the base *word2vec* encoder, extended the functionality to include the UNK tokens and padding, preprocessing and led the writing and reporting.

Tumelo designed and explored the subtopic on kernel-span choices, running and consolidating the comparative results.

Amaan Hanslod implemented the TextCNN classifier and end-to-end training/evaluation pipeline for the subtopic experiments.